Designing a Gearbox:
Shafts, Gears & Journals

MEGR 2156

Questions: Spend 5 minutes answering

Prior Knowledge:

What do you already know about gears and how they work to transmit motion or power?

Points of Confusion:

What aspects of gears or gear trains are unclear or confusing to you (e.g., tooth geometry, gear ratios, or contact forces)?

Applications:

Where do you see gears being used in real-world applications? Think beyond vehicles, what other systems or products rely on gears?

Design Perspective:

When designing or selecting gears, what factors do you think are most important to consider (e.g., torque, speed, material, alignment, or type of gear)?

Analysis and Selection:

Why is it important for engineers to analyze the loads, stresses, and deflections in gear teeth when designing a gear system?

Learning Objectives

- Design the material and geometry of a rotating shaft under torsion
- Gain an understanding the loads on a gear train: power, torque, force,speed
- Apply parameters to design a journal bearing

Design Shafts (Imperial Units)

What should be the diameter (inches) of a **power-transmission** to transmit 10 horsepower if the shaft is to make 150 revolutions per minute? Assume shaft's shear yield strength is 4000 psi. For **power-transmission** this includes the safety factor and the fatigue limit strength. Use equations from Machinery's Handbook on page 289. **Be careful of units.**

Parameters

P => Power [HP]

D => Diameter [inches]

N => Rotational Speed [rpm]

For main power-transmitting shafts:

$$P = \frac{D^3 N}{80} \quad (4a) \quad \text{or} \quad D = \sqrt[3]{\frac{80P}{N}} \quad (4b)$$

For lineshafts carrying pulleys:

$$P = \frac{D^3 N}{53.5} \quad (5a) \quad \text{or} \quad D = \sqrt[3]{\frac{53.5P}{N}} \quad (5b)$$

For small, short shafts:

$$P = \frac{D^3 N}{38} \quad (6a) \quad \text{or} \quad D = \sqrt[3]{\frac{38P}{N}} \quad (6b)$$

Design Shafts (SI units)

What should be the diameter of a **power-transmitting shaft** to transmit 150kW at 500 rpm? Assume shaft's shear yield strength is 28 MPa. For **power-transmission** this includes the safety factor and the fatigue limit strength. Use equations from Machinery's Handbook on page 291.

Parameters

P => Power [kW]

D => Diameter [mm]

N => Rotational Speed [rpm]

For main power-transmitting shafts:

$$P = \frac{D^3 N}{1.77 \times 10^6} \quad (10a) \qquad \text{or} \qquad D = \sqrt[3]{\frac{1.77 \times 10^6 P}{N}} \quad (10b)$$

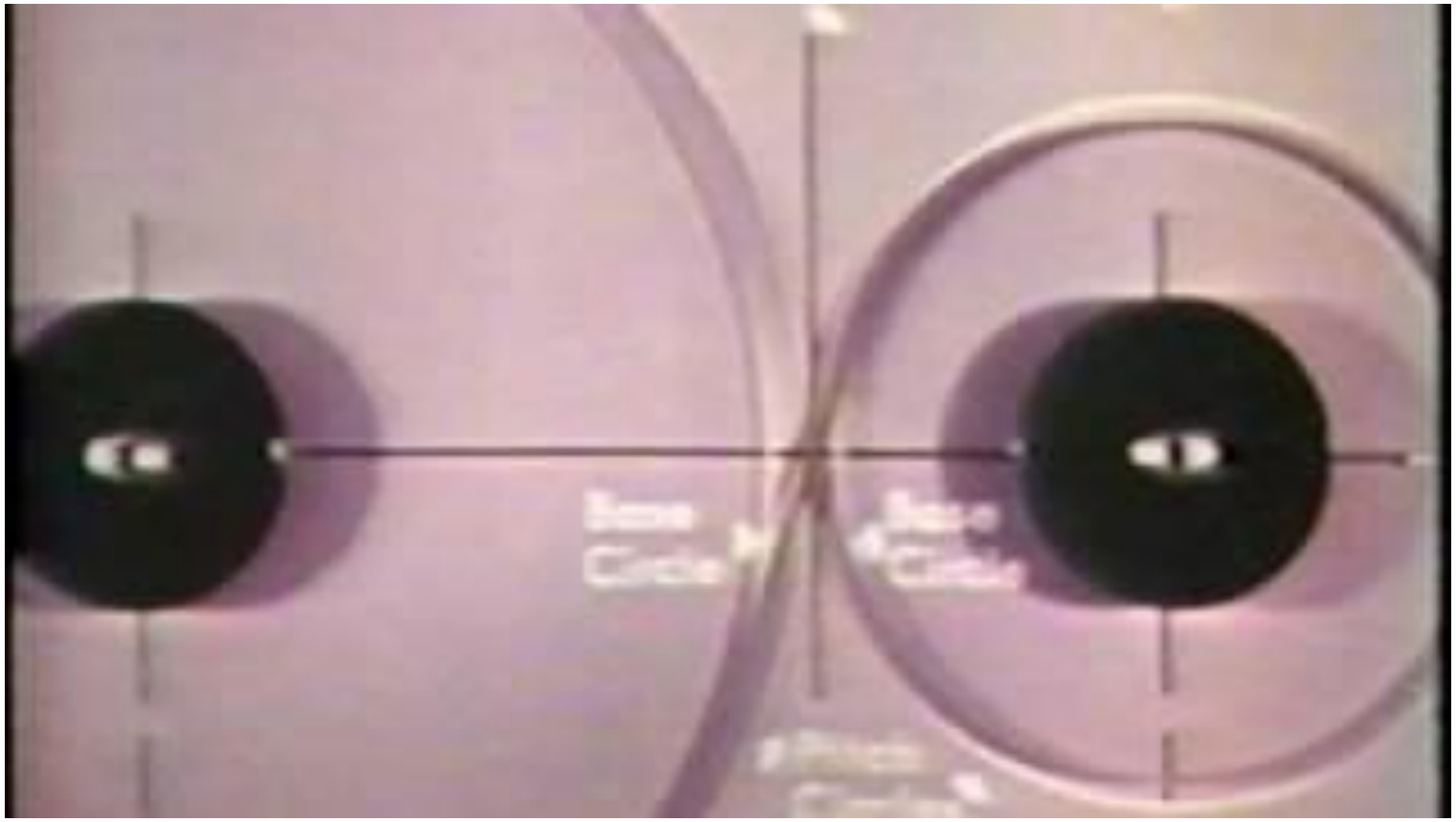
For small, short shafts:

$$P = \frac{D^3 N}{0.83 \times 10^6} \quad (11a) \qquad \text{or} \qquad D = \sqrt[3]{\frac{0.83 \times 10^6 P}{N}} \quad (11b)$$

Design Shafts

** What power would a **short shaft**, 50 millimeters in diameter, transmit at 400 rpm? Assume shaft's shear yield strength is 59 MPa. For short shaft this includes the safety factor and the fatigue limit strength. Use equations from Machinery's Handbook on page 297.

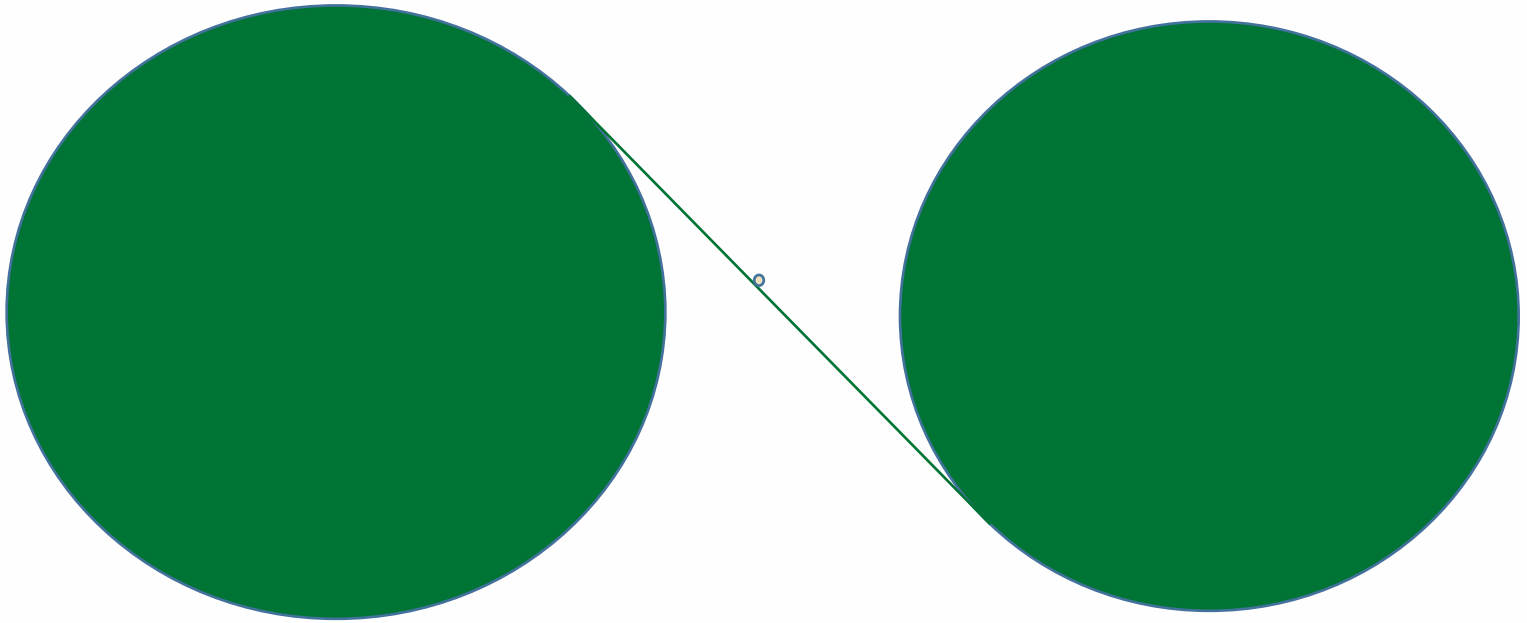
Fellows Gear Shaper: The Art of Generating Gears



After video go to Gear Force Analysis

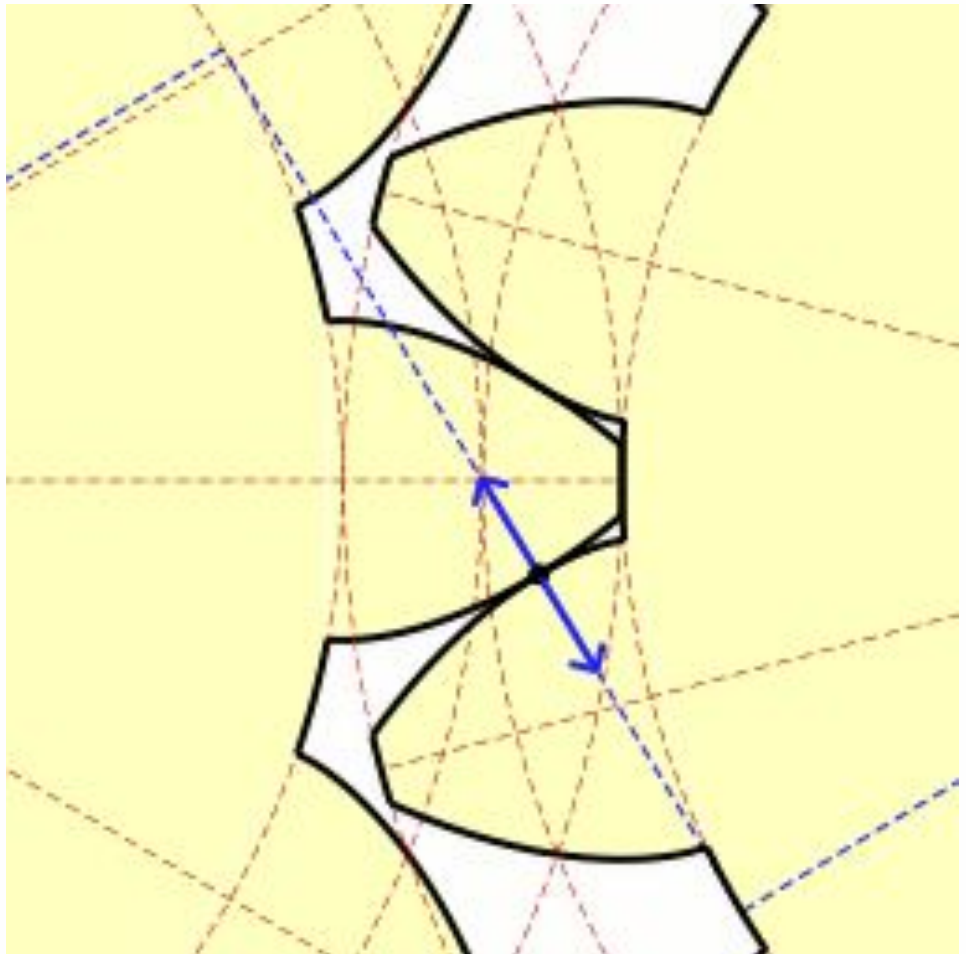
For power transfer, one desires smooth motion

- Ratio of angular motion of drive and driven shafts are constant
- A rope connecting smooth drums satisfies this requirement
- Consider a spot on the rope



The “spot” on the cable maps out an involute gear tooth form

- Provides constant speed
- Rolling, not sliding contact at the interface



Animated .gif courtesy Claudio Rocchini

Gear Nomenclature

Gears are referred to by **pitch diameter**

- Pitch diameter is measured at the point where the line of action intersects the line connecting the gear centers

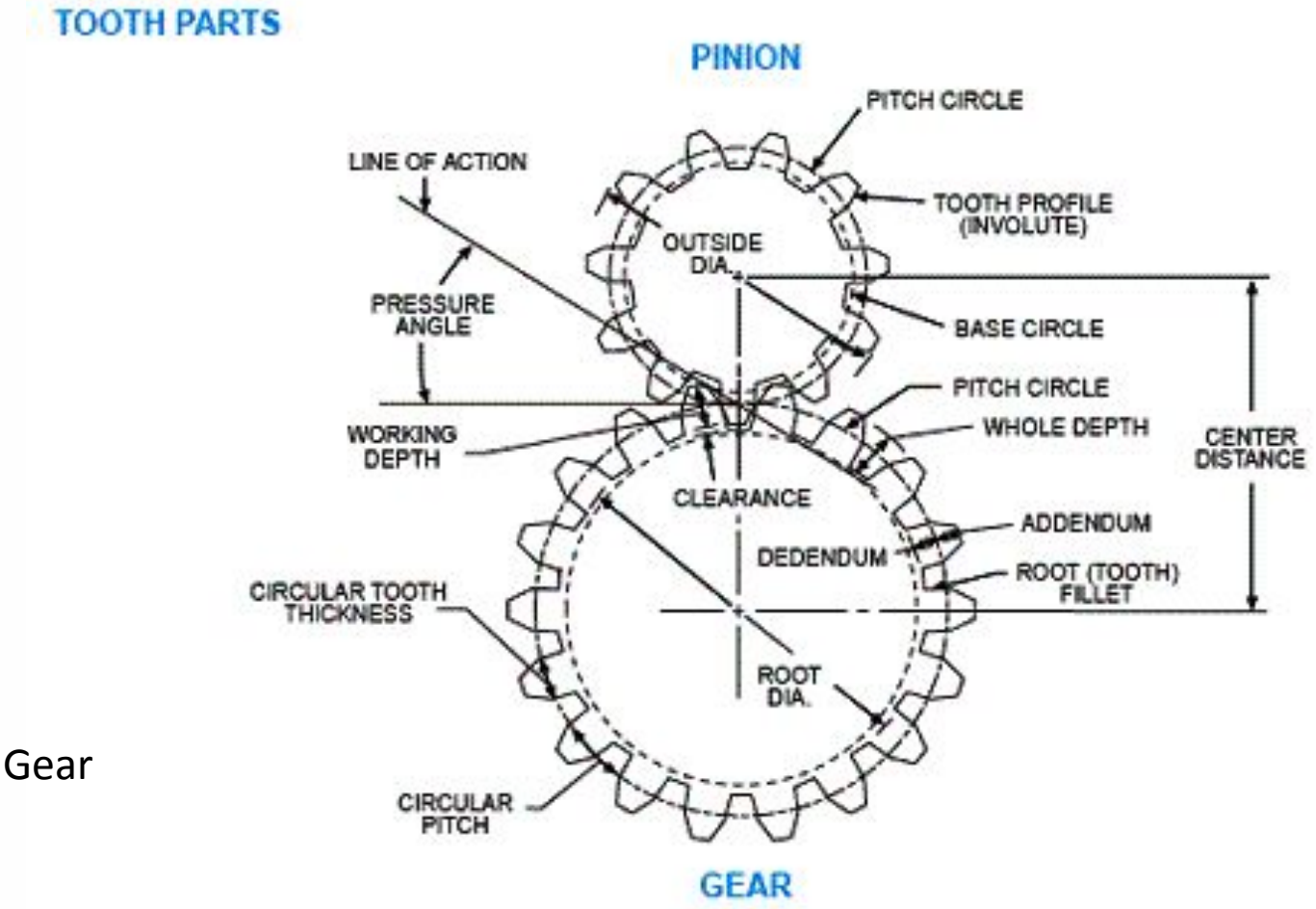


Image courtesy Boston Gear – also see MH p2129

Gear Nomenclature

Gears are referred to by **diametral pitch** and **pressure angle**

- **Diametral pitch** is the number of teeth *per inch of pitch diameter*
- **Pressure angle** is the angle between the line of action and the normal to the line connecting the centers of the gears

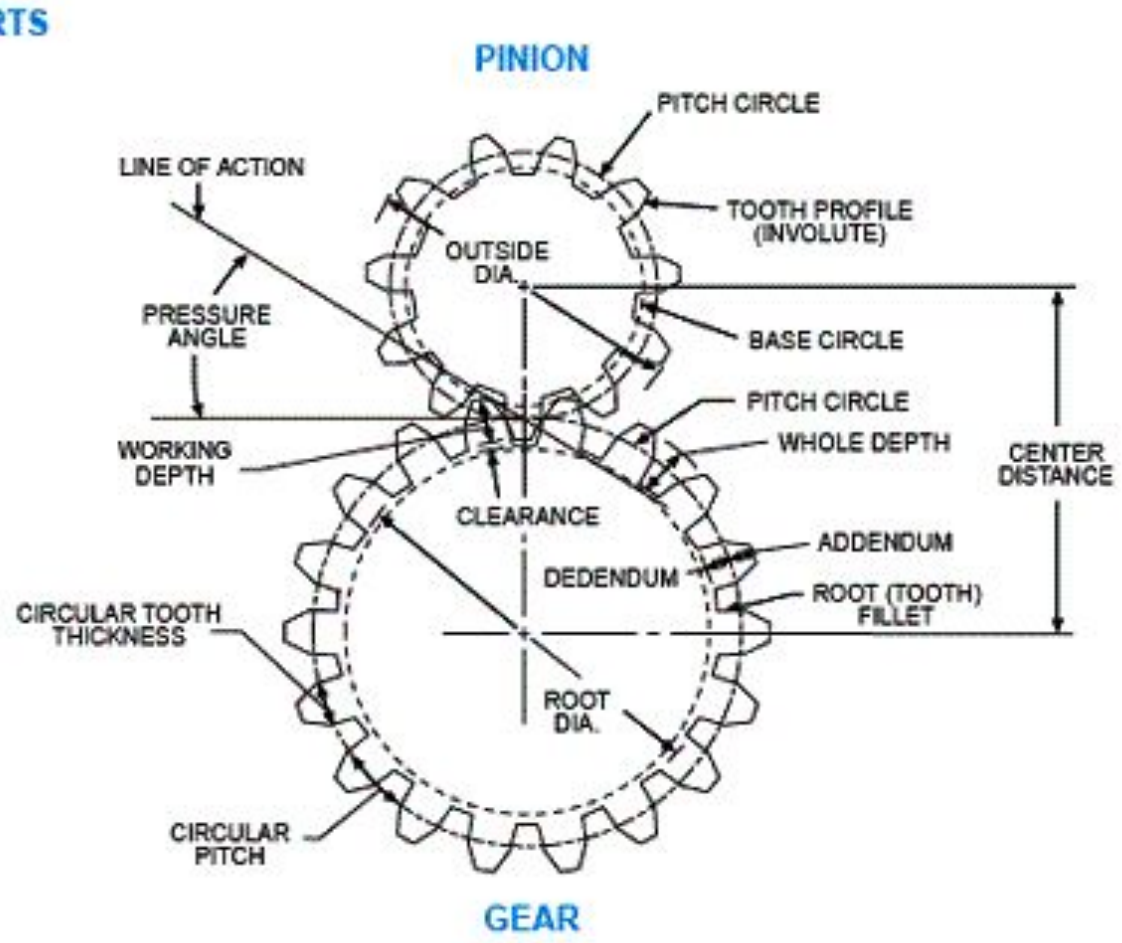


Image courtesy Boston Gear – also see MH p2129

Gears: Force Analysis

Transmission of torque through gears results in a radial force pushing them apart

$F_r = F_t (\tan(\alpha))$ & $F_t = 2T/d$

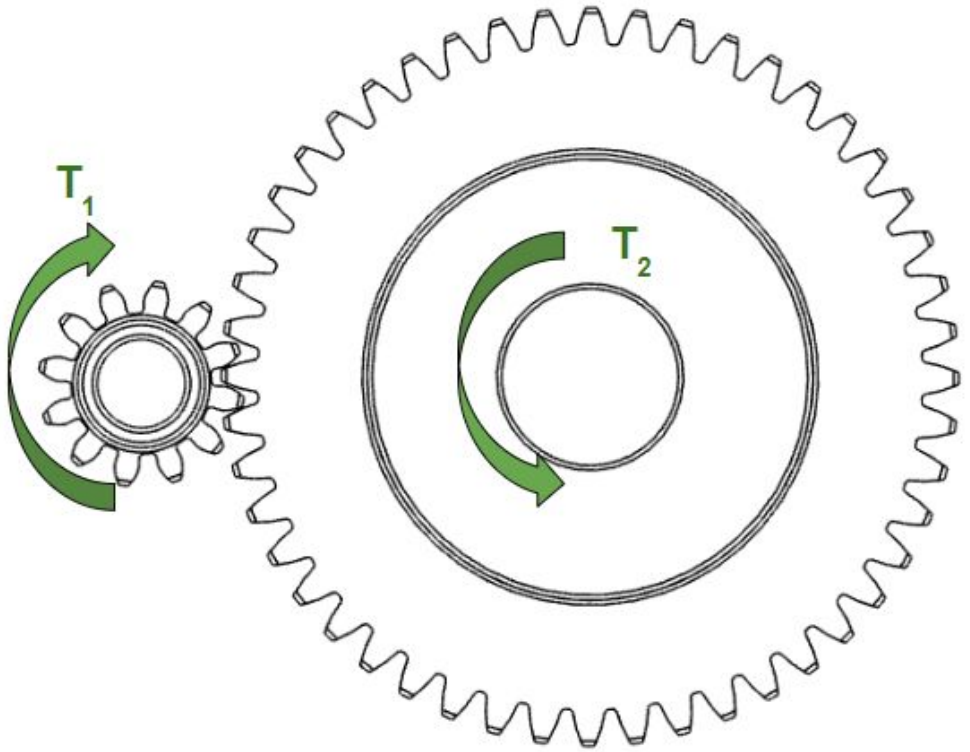
F_r = radial force

α = pressure angle

F_t = tangential force

T = torque

d = pitch diameter



Design the bearing with radial force and rotational speed due to the structural loop connecting the gear centers which should be stiff and the center distance precise.

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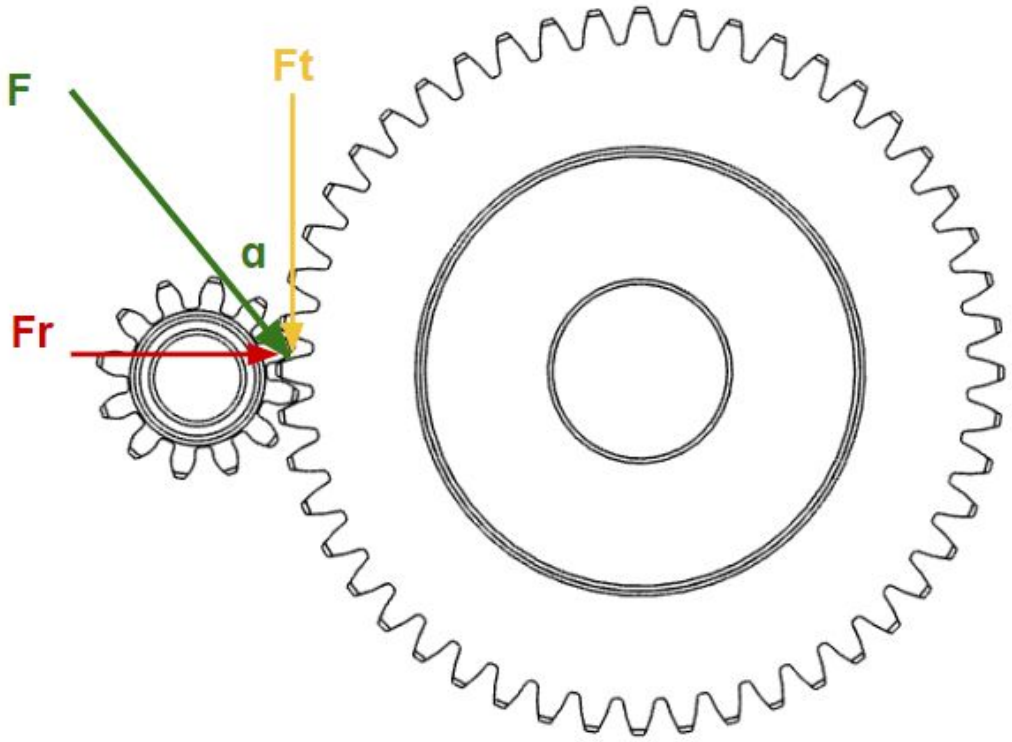
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Extra: Gear Teeth: Stress Analysis Overview

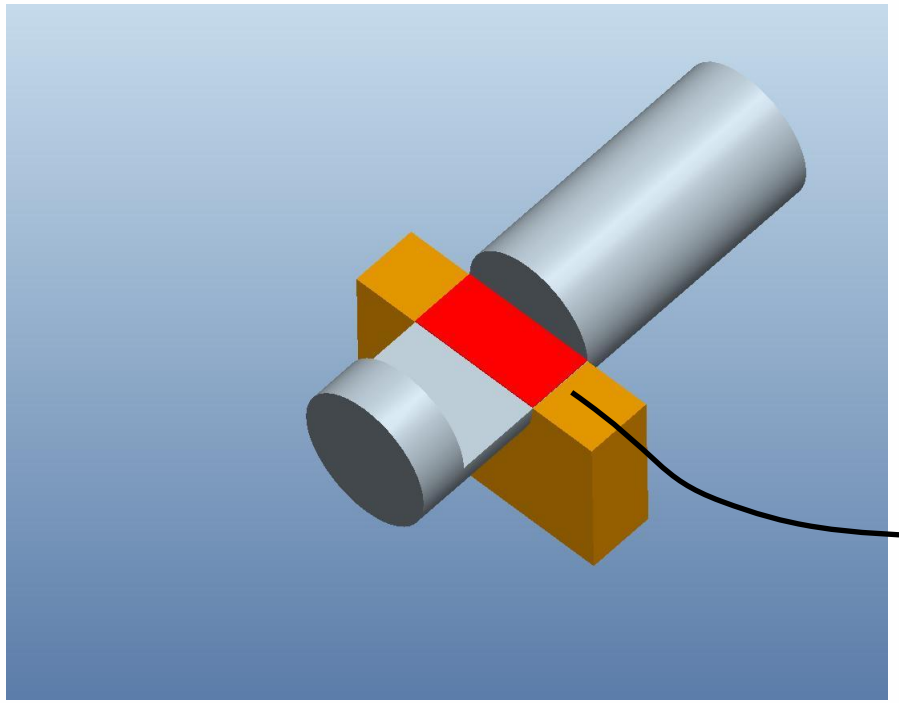
Torque limits of a gear is determined by the bending strength of the teeth

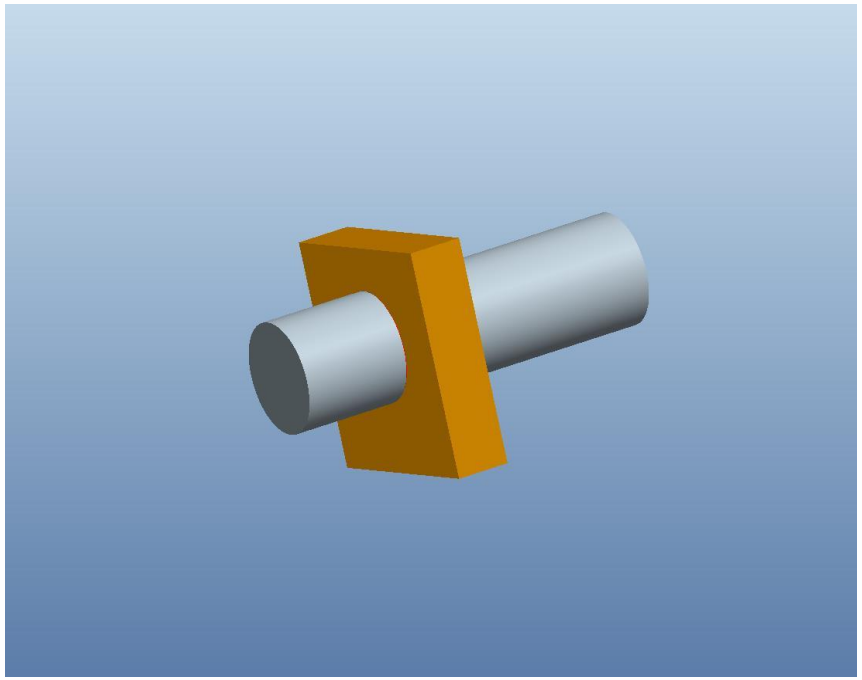
- Maximum moment occurs when tangential force is at maximum diameter
- Stress concentration at the root of the tooth must be considered
- Fatigue strength is the appropriate criterion since the load is cyclic

"Lewis Formula" is often used to estimate gear strength.

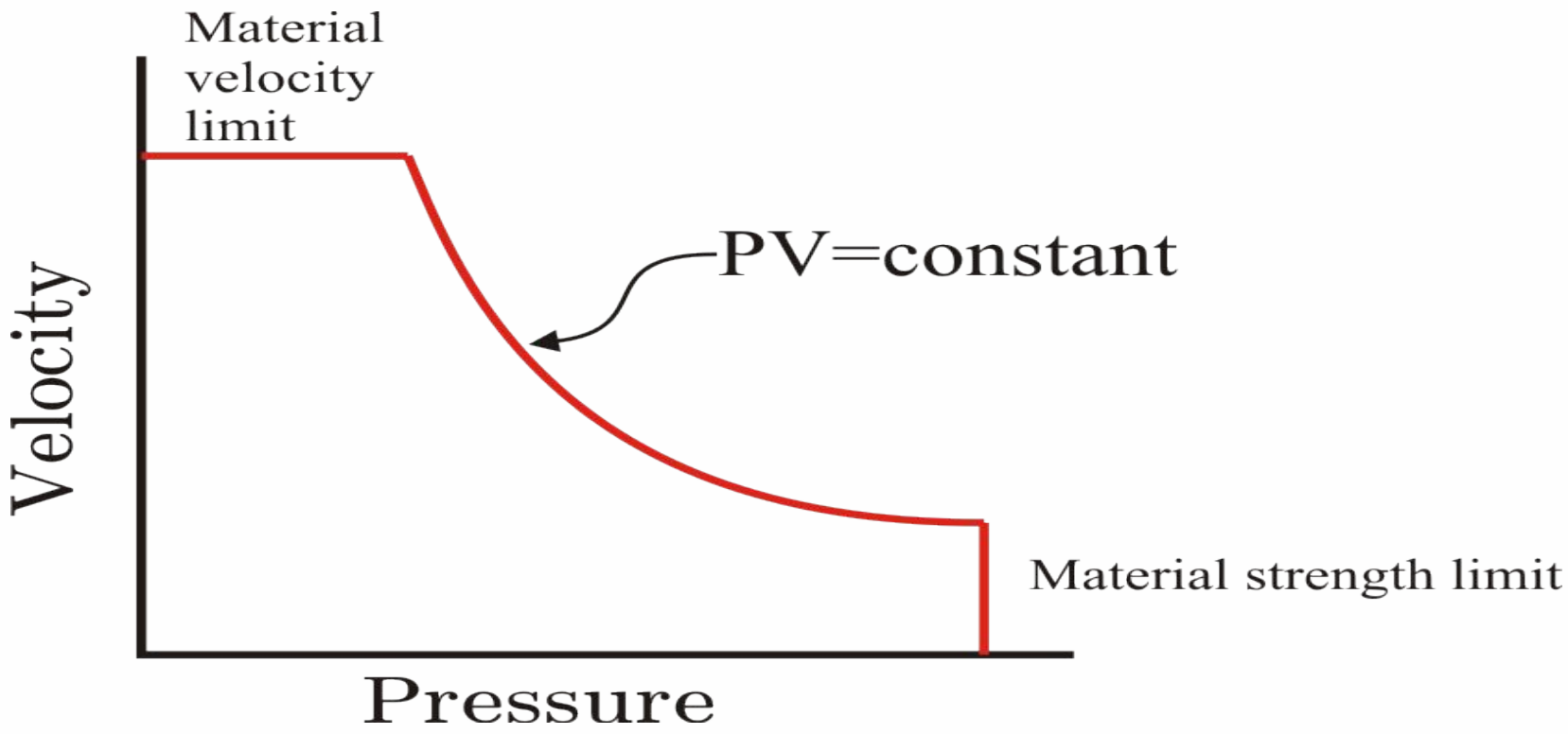
Calculating P and V for a journal bearing

Use the projected area and the applied load to calculate P for a journal bearing.





Typical Journal Bearing Safe Use



Allowable P and V values in a design are primarily a function of material and lubrication, and not specific bearing design.

16

Typical Journal Bearing Safe Use

Bearing material limits- need to look at all three for now.

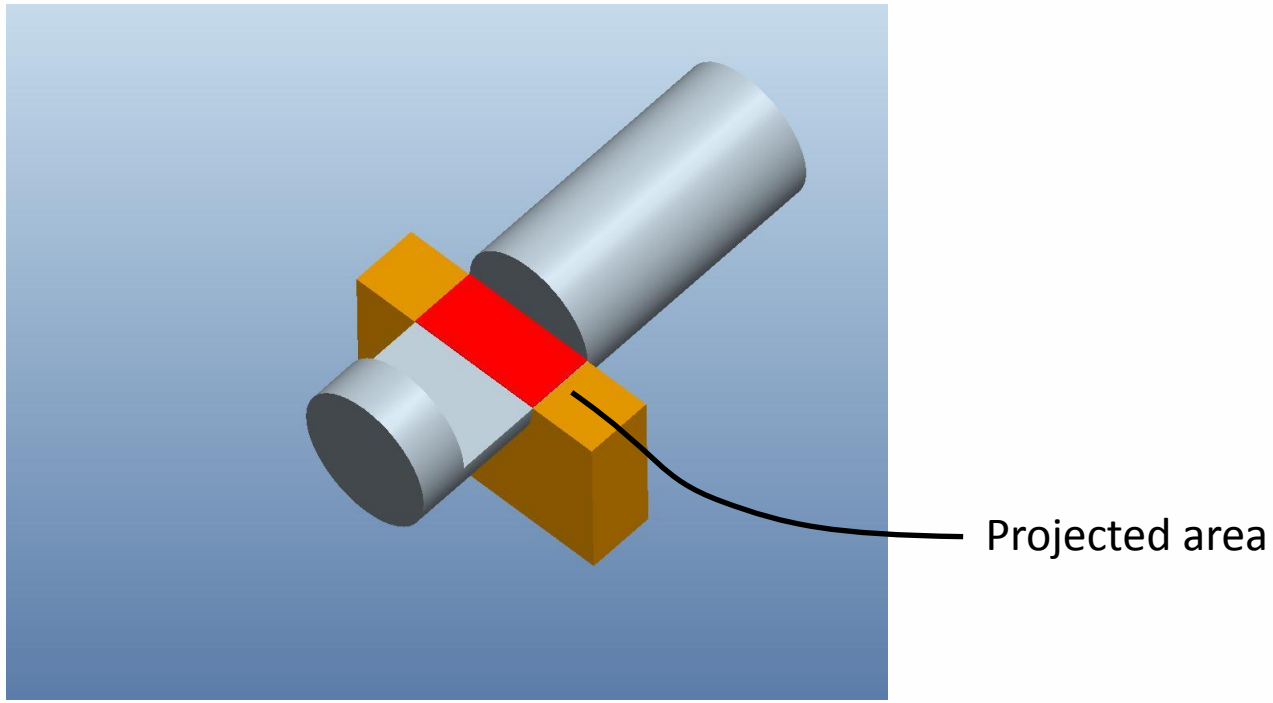
Table 6. Application Limits — Sintered Metal and Nonmetallic Bearings

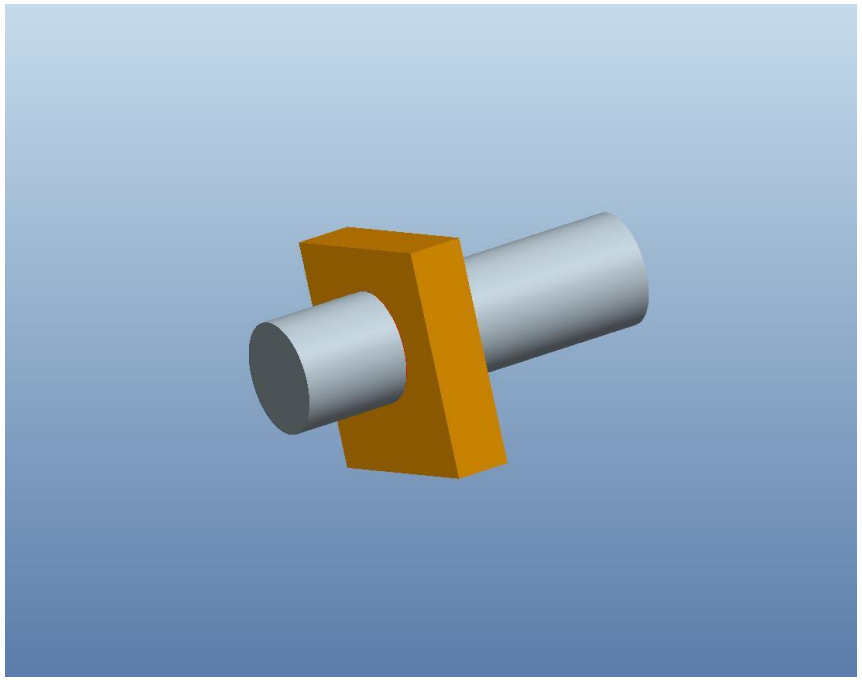
Bearing Material	Load Capacity (psi)	Maximum Temperature (°F)	Surface Speed, V_{max} (max. fpm)	PV Limit P = psi load V = surface ft/min
Acetal	1000	180	1000	3000
Graphite (dry)	600	750	2500	15,000
Graphite (lubricated)	600	750	2500	150,000
Nylon, Polycarbonate	1000	200	1000	3000
Nylon composite	...	400	...	16,000
Phenolics	6000	200	2500	15,000
Porous bronze	4500	160	1500	50,000
Porous iron	8000	160	800	50,000
Porous metals	4000–8000	150	1500	50,000
Virgin Teflon (TFE)	500	500	50	1000
Reinforced Teflon	2500	500	1000	10,000–15,000
TFE fabric	60,000	500	150	25,000
Rubber	50	150	4000	15,000
Maple & Lignum Vitae	2000	150	2000	15,000

Machinery's Handbook, 29th edition, page 2361

Example: Selecting Material for a Journal Bearing

One inch diameter shaft running in a half inch long journal carrying a 10 lb load at 5,000 rpm.





18

Example: Selecting Material for a Journal Bearing

One inch diameter shaft running in a half inch long journal carrying a 10 lb load at 5,000 rpm.

$P = F/A$

$F = 10$ lbf

$A = (.5 * 1)\text{in}^2$

$P = 20$ psi

19

Machinery's Handbook- Bearing Materials

Using the table below and the three criteria calculated you can choose a suitable material

P = 20 psi
V = 1309 fpm
PV = 26180 psi-fpm
The bearing material must be less than than the calculated numbers.

For instance, **Acetal** will not work due to the PV is less than calculated.

Table 4. Application Limits — Sintered Metal and Nonmetallic Bearings

Bearing Material	Load Capacity		Maximum Temperature		Surface Speed, V_{max} (max. fpm)	PV Limit P = psi load V = surface ft/min
	psi	kPa	°F	°C		
Acetal	1000	6895	180	82	1000	3000
Graphite (dry)	600	4137	750	399	2500	15,000
Graphite (lubricated)	600	4137	750	399	2500	150,000
Nylon, Polycarbonate	1000	6895	200	93	1000	3000
Nylon composite	…	…	400	204	…	16,000
Phenolics	6000	41369	200	93	2500	15,000
Porous bronze	4500	31026	160	71	1500	50,000
Porous iron	8000	55158	160	71	800	50,000
Porous metals	4000–8000	27579–55158	150	66	1500	50,000
Virgin Teflon (TFE)	500	3447	500	260	50	1000
Reinforced Teflon	2500	17237	500	260	1000	10,000–15,000
TFE fabric	60,000	413685	500	260	150	25,000
Rubber	50	345	150	66	4000	15,000
Maple & Lignum Vitae	2000	13790	150	66	2000	15,000

1 fpm = 0.3048 m/min; 1 psi = 6.8947 kPa;

20

Machinery's Handbook Bearing Materials

Using the table below and the three criteria calculated you can choose a suitable material

P = 20 psi
V = 1309 fpm
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Another instance that will work, Graphite (lubricated) because the calculated P, V, and PV are less the values in the table.

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21

Rules of Thumb for Bearing Materials

- Bearing surfaces are generally better if they are of dissimilar material
- Steel and aluminum are a bad combination
- Steel and anodized aluminum are OK
- Bearing dimensions should be precise enough to avoid “hot spots”
- Lubrication is important, but . . .
  - Do not oil nylon or most plastics

22

Some Solid Bearing Options



Porous bronze bushing



Rulon bushing



PTFE-lined bushing

23